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#6A:2

a) $Q = 18\ ^{g}/min$; $D_i = 3.068$ in

$l_{tot} = 95 ft$; 2 elbows @ $45° = 15D$

H_2O @ $68°F$: $\rho = 0.9982\ ^{g}/mL$

$\mu = 1.002 cP = 1.002 \times 10^{-3}\ Pa/s$

elevation @ 50 ft = 45°; $p_L = p_{atm} = 101325$

$D = 3.068 in\ (2.54 \times 10^{-2}\ ^{m}/in) \approx 0.0779 m$

$\langle v \rangle = Q / \pi (D/2)^2 \rightarrow Q = \dfrac{18(0.00378541\ m^3/s)}{60}$

$Q = 0.001136 \rightarrow \langle v \rangle = \dfrac{0.001136}{\pi (0.0779/2\ m)^2} \rightarrow$

$\langle v \rangle \approx 0.2383\ m/s$

Reynolds # - $Re = D\rho\langle v \rangle / \mu \rightarrow$

$$\frac{(0.0779 m)(0.9982 \times 10^3\ kg/m^3)(0.2383\ m/s)}{(1.002 \times 10^{-3}\ Pas)}$$

$Re = 1.85 \times 10^4$

flow = turbulent $\rightarrow$ smooth pipe ($k/D = 0$)

$f(Re = 1.85 \times 10^4, k/D = 0) \approx 0.0066$

- pipe length : 95ft

2 elbows @ 45° = 150

$L = [95\,(12+2)\,15\,(3.068)]\,in \cdot (2.54 \times 10^{-2}\ m/in)$

$L = 31.29\ m$

- pressure drop :

fanning form $- f = 1/4\,(D/L)\left(\mathcal{P}_0 - \mathcal{P}_L / 1/2\,\rho\langle v\rangle^2\right) \rightarrow$

$$\mathcal{P}_0 - \mathcal{P}_L = \frac{2fL\rho\langle v\rangle^2}{D} \rightarrow$$

$$= \frac{2\,(0.0066 \cdot 31.29\,m \cdot (0.9982 \times 10^3\ kg/m^3) \cdot 0.238}{0.0779\ m}$$

$\mathcal{P}_0 - \mathcal{P}_L = 3.01 \times 10^2\ Pa \rightarrow \Delta\mathcal{P}_f = 300.6\ Pa$

$p_0 - p_L + \rho(\Phi_0 - \Phi_L) = 300.6\ Pa$; $p_L = 101325\ Pa$

$\hookrightarrow \rho g(z_0 - z_L) = -\rho g \Delta z$

$\Delta z = 50ft\,\sin(45°) \rightarrow 50\,(0.3048)\sin(45°)$

$\Delta z = 10.77m$

$-\rho g \Delta z \rightarrow = -0.9982 \times 10^3 (9.81)(10.77)$

$= -105463.52\ Pa$

$p_0 - p_L + \rho(\Phi_0 - \Phi_L) = 300.6\ Pa \rightarrow$

$p_0 = 300.6\,Pa + p_L - \rho(\Phi_0 - \Phi_L) \rightarrow$

$p_0 = 300.6\,Pa + 101325\,Pa + 105463.52\,Pa$

$p_0 = 207089.12\,Pa \rightarrow$

$p_{o,gauge} = p_o - p_{atm} \rightarrow = 207089.12 - 101325$

$p_{o,gauge} = 105764.12 \text{ Pa}$

$p_{o,gauge} = \dfrac{300.6 + 105463.52}{6894.76} \text{ psig}$

$p_{o,gauge} = 15.34 \text{ psig}$

b) $\text{fraction} = \dfrac{(p_o - p_L)_{frict}}{(p_o - p_L)_{total}} = \dfrac{300.6 \text{Pa}}{105463.52 \text{Pa}}$

$\text{fraction} = 0.285\%$

#6A.3:

a) $\Delta P = {}^{2fL}/_{D} (\rho \langle v \rangle^2) \rightarrow f = {}^{D\Delta P}/_{2L\rho\langle v\rangle^2}$

turbulent flow: $1/\sqrt{f} = -1.56 \log_{10}\left[{}^{6.9}/_{Re} + {}^{(k/D)}/_{3.7}\right]^{10/9}$

$k/D = 0 \rightarrow 1/\sqrt{f} = -1.56 \log_{10}({}^{6.9\mu}/_{D\rho\langle v\rangle})$

$\hookrightarrow \sqrt{{}^{2L\rho}/_{D\Delta P}}\langle v\rangle$

$\sqrt{{}^{2L\rho}/_{D\Delta P}}\langle v\rangle = -1.56 \log_{10}({}^{6.9\mu}/_{D\rho\langle v\rangle})$

$\hookrightarrow \sqrt{\dfrac{2(402.3)(0.9982 \times 10^3)}{0.1524(1724)}} = 55.29$

${}^{6.9\mu}/_{D\rho} = \dfrac{6.9(1.002 \times 10^{-3})}{0.1524 \cdot (0.9982 \times 10^3)} = 4.55 \times 10^{-5}$

$$5.53\langle v\rangle = -1.56\log_{10}\left(4.55\times10^{-5}/\langle v\rangle\right)$$

Excel to solve for $\langle v\rangle$ & Q:

$\langle v\rangle = 0.243\ m/s$

solving for Q:

pipe radius $(R = D/2) = 0.0762$

flow rate $(Q = \pi R^2 u) = \pi(0.0762)^2(0.243)$

$$Q = 0.004428$$

$$0.004428(264.172)(3600) = 4210.737\ gal/hr$$

initial guess	0.242722
constant	55.3
constant	1.56
constant	4.45E-05
LHS	1.34E+01
RHS	1.34E+01
Residual	-1.02E-06
Q	
pipe radius	0.0762
flow rate	0.004428
convert	4210.737

#6A.9

$D = 4 in \rightarrow 4(2.54 \times 10^{-2}) = 0.1016 m$

$R = D/2 \rightarrow 0.1016/2 = 0.0508 m$

$L = 5.5 ft \rightarrow 5.5(0.3048) = 1.6764 m$

$D_p = 1/16 in \rightarrow 0.0625(2.54 \times 10^{-2}) = 0.0016 m$

porosity : $\varepsilon = 0.41$

$\mu = 1.495 \times 10^{-5}$ Pas

avg gas density : $\rho = PM/RT \rightarrow$

inlet - $\rho_0 = \dfrac{25 \cdot 101325 \cdot (44 \times 10^{-3})}{8.314 \cdot 300} = 44.7 \, kg/m^3$

outlet - $\rho_2 = \dfrac{3 \cdot 101325 \cdot (44 \times 10^{-3})}{8.314 \cdot 300} = 5.36 \, kg/m^3$

$\bar{\rho} = (\rho_0 + \rho_2 / 2) \rightarrow \dfrac{44.7 + 5.36 \, kg/m^3}{2} \rightarrow$

$\bar{\rho} = 25.03 \, kg/m^3$

- turbulent flow : $Re_h = \dfrac{2 D_p G_0}{3 \mu (1-\varepsilon)} > 650 \rightarrow$

$$\Delta P / L = \frac{150(1-\varepsilon)^2}{\varepsilon^3} \cdot \frac{\mu G_0}{\bar{\rho} D_p^2} + \frac{1.75(1-\varepsilon)}{\varepsilon^3} \cdot \frac{G_0^2}{\bar{\rho} D_p}$$

high turbulence

$$\Delta p / L = \frac{1.75(1-\varepsilon)}{\varepsilon^3} \cdot \frac{G_0^2}{\bar{\rho} D_p^2} \rightarrow$$

$$\frac{p_0 - p_L}{L} = 7/4 \left(\frac{1-\varepsilon}{\varepsilon^3} \cdot \frac{G_0^2}{\bar{\rho} D_p} \right) \rightarrow$$

$$G_0 = \sqrt{\frac{\Delta p}{L} \cdot \frac{\varepsilon^3 \bar{\rho} D_p}{7/4(1-\varepsilon)}}$$

$$G_0 = \sqrt{\frac{2.23 \times 10^6}{1.68} \cdot \frac{(0.41)^3 (25.03)(1.59 \times 10^{-3})}{7/4(1-0.41)}}$$

$$G_0 = 59.38 \ \text{kg}/\text{m}^2\text{s}$$

mass flow rate: $w = G_0 A \rightarrow = G_0 \pi R^2$ $\overset{=A}{}$

$$w = 59.38 \ \text{kg}/\text{m}^2\text{s} \cdot \pi \cdot (0.0508 \ \text{m})^2$$

$$w = 0.481 \ \text{kg}/\text{s} = 481.41 \ \text{g}/\text{s}$$

check: $Re_n = \frac{2 D_p G_0}{3\mu(1-\varepsilon)} \rightarrow \frac{2(1.59 \times 10^{-3})(59.4)}{3(1.495 \times 10^{-5})(1-0.41)} \rightarrow$

$\checkmark$ $Re_n = 7138.37 > 650$ = high turbulence

#6B.2:

a) laminar

$$F_k = 1.328\,W(\rho\mu v_\infty^3 L)^{1/2} \rightarrow = 1.328(\rho\mu L W^2 v_\infty^3)^{1/2}$$

friction factor $\rightarrow F_k = 2LW(\tfrac{1}{2}\rho v_\infty^2 f)$

$$1.328(\rho\mu L W^2 v_\infty^3)^{1/2} = 2LW(\tfrac{1}{2}\rho v_\infty^2 f)$$

$$f = \frac{1.328(\rho\mu L W^2 v_\infty^3)^{1/2}}{\rho v_\infty^2 L W} \rightarrow$$

$$= 1.328\,\frac{\sqrt{\rho\mu L v_\infty^3}}{\rho v_\infty^2 L} \rightarrow = 1.328\sqrt{\frac{\mu}{L\rho v_\infty}} = f$$

Reynolds #: $Re = \rho v_\infty L/\mu \rightarrow = \mu/L\rho v_\infty = 1/Re$

$$f = 1.328\sqrt{1/Re} \rightarrow f = \frac{1.328}{\sqrt{Re}}$$

b) $F_k = 0.074\,\rho v_\infty^2 WL(Lv_\infty\rho/\mu)^{-1/5}$

friction factor: $F_k = 2LW(\tfrac{1}{2}\rho v_\infty^2 f)$

$$0.074\,\rho v_\infty^2 WL(Lv_\infty\rho/\mu)^{-1/5} = 2LW(\tfrac{1}{2}\rho v_\infty^2 f)$$

cancel $\rightarrow 0.074\,(Lv_\infty\rho/\mu)^{-1/5} = f$

Reynolds #: $Re = (Lv_\infty\rho/\mu) \rightarrow$

$f = 0.074(Re)^{-1/5}$ for $5\times10^5 < Re < 2\times10^7$

#6B.3:

$$v_z(x) = \Delta P/2\mu L\,(x^2 - B^2) \;;\; \Delta P = p_0 - p_L$$

shear stress @ $x = B$

$$\tau_{zx}|_{x=B} = -\mu\,(dv_z/dx)|_{x=B} \rightarrow = \mu\left[\Delta P/\mu L \cdot x\right]_{x=B} \rightarrow$$
$$= -\Delta P/L \cdot B$$

tot. drag force on both plates $\rightarrow F_k = 2LW\,\tau_{zx}|_{x=B}$

avg. velocity: $\langle v_z\rangle = 1/2B \int_{-B}^{B} v_z(x)\,dx \rightarrow$

$$= 1/2B \int_{-B}^{B} \Delta P/2\mu L\,(x^2 - B^2)\,dx \rightarrow \Delta P/2\mu L \text{ out} \rightarrow$$

$$\int_{-B}^{B} (x^2 - B^2)\,dx \rightarrow = \left[x^3/3 - B^2 x\right]_{-B}^{B} \rightarrow$$

$$(B^3/3 - B^3) - (-B^3/3 + B^3) = -4B^3/3$$

$$\langle v_z\rangle = 1/2B\,(\Delta P/2\mu L)(-4B^3/3) \rightarrow = -\Delta P B^2/3\mu L \rightarrow$$

$$\langle v_z\rangle = \Delta P B^2/3\mu L$$

wall shear w/ $\langle v_z\rangle \rightarrow |\tau_w| = \Delta P/L \cdot B \rightarrow$

$$= 3\mu(\langle v_z\rangle/B)$$

friction factor:

$$f = F_k/2LW(1/2\rho\langle v_z\rangle^2) = \tau_w/1/2\rho\langle v_z\rangle^2 \rightarrow$$

$$= \frac{3\mu(\langle v_z\rangle/B)}{(1/2\rho\langle v_z\rangle^2)} = \frac{6\mu}{B\rho\langle v_z\rangle}$$

Reynolds #: $Re = 2B\rho\langle v_z\rangle/\mu$

$$\mu/B\rho\langle v_z\rangle = 2/Re \rightarrow$$

$$f = 6(\mu/B\rho\langle v_z\rangle) \rightarrow f = 6(2/Re) \rightarrow f = 12/Re$$

mean hydraulic radius: laminar flow

$f = 16/Re_h \rightarrow Re_h = 4R_h\rho\langle v\rangle/\mu$;

$R_h = A/P_w$; $A = 2BW$; $P_w = 2(2B+W)$

$R_h = 2BW/2(2B+W) = BW/2B+W \rightarrow$

narrow slit $\rightarrow R_h = BW/B = B$

$Re_h = 4R_h\rho\langle v_z\rangle/\mu \rightarrow 4B\rho\langle v_z\rangle/\mu$

$$f = 16/Re_h \rightarrow \frac{16}{4B\rho\langle v_z\rangle/\mu} \rightarrow = \frac{4\mu}{B\rho\langle v_z\rangle}$$

$Re = 2B\rho\langle v_z\rangle/\mu \leftarrow f = 8/2B\rho\langle v_z\rangle/\mu$

$f_{MHR} = 8/Re$

compare $8/Re$; $12/Re \rightarrow 12/Re / 8/Re = 12/8$

#6B.5:

a) $f = 0.0791/Re^{1/4}$ → $Re = \rho\langle v\rangle D/\mu$

mass flow: $w = \rho A\langle v\rangle = \rho\left(\pi(D')^2/4\right)\langle v\rangle'$ →

$\rho(\pi D^2/4)\langle v\rangle = \rho\left(\pi(D')^2/4\right)\langle v\rangle'$ →

$D^2\langle v\rangle = (D')^2\langle v\rangle'$ ∴ $D' = D/2$ →

$D^2\langle v\rangle = (D/2)^2\langle v\rangle'$ → $= D^2/4\,\langle v\rangle'$ →

$\langle v\rangle' = 4\langle v\rangle$

new Re #: $Re' = \rho\langle v\rangle' D'/\mu$ → $= \dfrac{\rho(4\langle v\rangle)(D/2)}{\mu}$ →

$2(\rho\langle v\rangle D/\mu)$ → $= 2Re = Re'$

new f #: $f' = 0.0791/(2Re')^{1/4}$ → $2^{-1/4}\left(0.0791/Re^{1/4}\right)$ →

$f' = f/2^{1/4}$

pressure drop to friction factor:

$f = D/4L\left(\Delta P/\tfrac{1}{2}\rho\langle v\rangle^2\right)$ → $\Delta P = 2f(L/D)\rho\langle v\rangle^2$ →

new pressure drop: $\Delta P' = 2f'(L/D')\rho\langle v\rangle'^2$

$\Delta P'/\Delta P$ → $\dfrac{f'}{f}\cdot\dfrac{D}{D'}\cdot\dfrac{\langle v\rangle'^2}{\langle v\rangle^2}$ → $= \left(\dfrac{1}{2^{1/4}}\right)\cdot 2\cdot 16 = \dfrac{32}{2^{1/4}}$

$\Delta P'/\Delta P = 32/2^{1/4} = 26.91$ → $\Delta P' = 26.91\Delta P$

b) og: $Re = \rho \langle v \rangle D / \mu$; $\Delta P = 2f(L/D)\rho \langle v \rangle^2$

new: $\langle v \rangle' = 4\langle v \rangle$; $D' = D/2$

$Re' = \rho \langle v \rangle' D' / \mu = 2Re$

$\Delta p' = 2f'(L/D')\rho \langle v \rangle'^2$

find ratio between f and f' with knowing $Re' = 2Re$.

#6B.7:

a) weight: $W = 4/3 \pi R^3 \rho_{sph} g$

drag force: $F_D = (\text{proj. area})(1/2 \rho v^2) f \rightarrow$

$= \pi R^2 (1/2 \rho v^2) f$; $f = 0.44$; $Re \geq 6000$

newton 2nd: weight - drag $= m(dv/dt)$

$m = 4/3 \pi R^3 \rho_{sph} \rightarrow$

$4/3 \pi R^3 \rho_{sph} g - \pi R^2 (1/2 \rho v^2) f = 4/3 \pi R^3 \rho_{sph} \, dv/dt \rightarrow$

$4/3 R \rho_{sph} g - 1/2 \rho f v^2 = 4/3 R \rho_{sph} \, dv/dt \rightarrow \div 4/3 R \ldots$

$1/g \, dv/dt = 1 - (3\rho f / 8R\rho_{sph} g) v^2 \rightarrow$

$c^2 = 3\rho f / 8R\rho_{sph} g$

ODE: $1/g \, dv/dt = 1 - c^2 v^2 \rightarrow dv/dt = g(1 - c^2 v^2)$

I.C: $v(0) = 0$ solve for $v(t) \rightarrow \dfrac{dv}{1 - c^2 v^2} = g\,dt$

$$\int_0^v dv'/1-c^2v'^2 = \int_0^t g\,dt' \rightarrow$$

$$1/c \tanh^{-1}(cv)\Big|_0^v = gt \rightarrow 1/c \tanh^{-1}(cv) = gt \rightarrow$$

$$\tanh(\tanh^{-1}(cv) = cgt) \rightarrow cv = \tanh(cgt) \rightarrow$$

$$v(t) = 1/c \tanh(cgt)$$

distance fall $(z(t))$:

$$dz/dt = v(t) = 1/c \tanh(cgt) \;\; \& \;\; I.C = z(0) = 0 \rightarrow$$

$$z(t) = \int_0^t v(t')dt' \rightarrow = 1/c \int_0^t \tanh(cgt')dt' \rightarrow u = cgt' \rightarrow$$

$$du = cg\,dt' \rightarrow dt' = du/cg \rightarrow$$

$$z(t) = 1/c \int_{u=0}^{u=cgt} \tanh u \; du/cg \rightarrow = 1/c^2g \int_0^{cgt} \tanh u\,du \rightarrow$$

$$\int \tanh(u)du = \ln(\cosh(u)) \rightarrow$$

$$z(t) = 1/c^2g \left[\ln(\cosh(u))\right]_0^{cgt} \rightarrow$$

$$1/c^2g\left[\ln(\cosh(cgt)) - \ln(\cosh(0))\right] \rightarrow$$

$$z(t) = 1/c^2g \ln[\cosh(cgt)]$$

b) accel $= 0$ so $dv/dt = 0$

$$4/3\pi R^3 \rho_{sph} g - \pi R^2 (1/2 \rho v_T^2) f = 0 \rightarrow$$

$$4/3\pi R^3 \rho_{sph} g = \pi R^2 (1/2 \rho v_T^2) f \rightarrow$$

$$4/3 R \rho_{sph} g = 1/2 \rho f v_T^2 \rightarrow v_T^2 = \frac{4/3 R \rho_{sph} g}{1/2 \rho f} \rightarrow$$

$$v_T^2 = 8/3 \left(R\rho_{sph} g/\rho f\right) \rightarrow$$

$$v_T = \sqrt{8/3 \left(R\rho_{sph} g/\rho f\right)} \rightarrow$$

$$c^2 = 3\rho f / 8R\rho_{sp} n g \rightarrow 1/c^2 = 8R\rho_{sp} n g / 3\rho f \rightarrow$$

$$v_T^2 = \frac{8R\rho_{sp} n g}{3\rho f} = 1/c^2 \rightarrow v_T = 1/c$$